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Simulation of Normally Off AlGaN/GaN HFET With P-Type GaN Gate and AlGaN Buffer

Abstract

This TCAD Sentaurus™ project provides a template setup for the simulation of DC characteristics of a normally off AlGaN/GaN HFET with p-type GaN gate and AlGaN buffer [1]. Polarization charges at interfaces, due to divergence of polarization vectors, are computed automatically using the built-in strain piezoelectric polarization model [2].

In the project, examples of interface trap placement are given, and appropriate models are selected for the simulation of III–nitride devices. Changes to the numeric accuracy settings are required to simulate large bandgap materials such as GaN and AlGaN.

I_d–V_g and I_g–V_g curves for V_d = 15 V, I_d–V_d curves for V_g = 1 V, 3 V, 5 V, and the off-state breakdown curve for V_d up to 1000 V are simulated.

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Introduction

The purpose of this simulation project is to provide a device simulation setup optimized for the simulation of a normally off p-gate GaN HFET.

The p-gate GaN HFET used in this project is based on the structure published in [1]. Mg-doped GaN is used as the p^+ gate to deplete the channel at $V_g = 0$, yielding a normally off transistor. An AlGaN buffer is used to increase the threshold voltage (V_{th}), while reducing the R_{on} with a higher Al content barrier. The AlGaN buffer also suppresses punch-through [1].

One of the main characteristics of III–nitride materials is the presence of polarization vectors, with spontaneous and piezoelectric components. The primary effect of polarization is an interface charge due to abrupt variations in the polarization at the AlGaN–GaN heterointerface and at the AlGaN surface. The resulting values are computed automatically using the built-in strain piezoelectric polarization model in Sentaurus Device [2].

The large band gap of III–nitride semiconductors requires some adjustments to the numeric accuracy settings of Sentaurus Device to ensure robust and accurate simulation results, which can be found in this template. In particular, to improve the convergence of the breakdown simulation, transient simulation is used with minute amounts of background carrier generation. Such an approach and specific numeric accuracy settings also can be used for other GaN-based devices.

It is assumed that users are familiar with the Sentaurus tool suite, in particular, Sentaurus Workbench, Sentaurus Structure Editor, Sentaurus Device, and Sentaurus Visual. For an introduction and tutorials, refer to the TCAD Sentaurus Tutorial. For details about tool uses and specific tool syntax, refer to the respective manuals.

General Simulation Setup

This section describes the tool flow of the Sentaurus Workbench project. For each tool, the associated Sentaurus Workbench input parameters are discussed.

Sentaurus Structure Editor

Sentaurus Structure Editor defines the p-gate GaN HFET structure. The Sentaurus Workbench parameters `FP`, `Lgd`, and `AlGaN_Doping` are used by Sentaurus Structure Editor, and they define the length of the field plate, the gate–drain distance, and the doping level of the top AlGaN barrier layer, respectively.

In the template, two device structures are simulated:

- Structure 1 has `FP = 1.8` and `Lgd = 6`.
- Structure 2 has `FP = 5` and `Lgd = 10`.

Both device structures have `AlGaN_Doping = 1e18` cm^{-3}.

Sentaurus Device

The three Sentaurus Device instances (`BV`, `IdVg`, and `IdVd`) perform the off-state breakdown, and the I_d–V_g and I_d–V_d sweeps, respectively.

The Sentaurus Workbench parameters `RScale` and `BScale` scale the parameter `Rinf` in the Schottky resistance model and the GaN parameters `b(low)` and `b(high)` in the van Overstraeten – de Man model so as to adjust the gate leakage current and breakdown voltage, respectively.

The Sentaurus Workbench parameters `Vd` and `Vg` define the constant V_d and V_g during the I_d–V_g and I_d–V_d sweeps, respectively. `Vgmax` and `Vdmax` define the maximum V_g and V_d during the I_d–V_g and I_d–V_d sweeps, respectively.

The sweeping conditions are:

- Off-state breakdown (BV) sweep from $V_d = 0$ V to 1000 V at $V_g = 0$ V
- I_d–V_g sweeps from V_g = `Vgmax` to –2 V at $V_d = 15$ V
- I_d–V_d sweeps from $V_d = 0$ V to `Vdmax` at $V_g = 1$ V, 3 V, or 5 V

Sentaurus Visual

The three Sentaurus Visual instances, corresponding to the three Sentaurus Device instances, plot the off-state BV, and the I_d–V_g and I_d–V_d curves.

Tool-Specific Setups

Device Generation Using Sentaurus Structure Editor and Sentaurus Mesh

Sentaurus Structure Editor defines the p-gate GaN HFET structure and the meshing criteria.

The simulated device structure is based on the paper [1]. Wherever the parameters are not given in the paper, reasonable parameters are used. Figure 1 on page 4 shows the cross section of the layers at the gate region. The structure includes a 2 µm $Al_{0.05}Ga_{0.95}N$ buffer layer, a 10 nm GaN channel layer, and a 15 nm $Al_{0.23}Ga_{0.77}N$ barrier layer. The gate is a 110 nm p-type GaN layer. The source–gate separation is 1 µm and the gate length is 1.4 µm.

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Highly doped regions are created under the source–drain electrodes down to the GaN channel to emulate metal spikes and to control contact resistance.

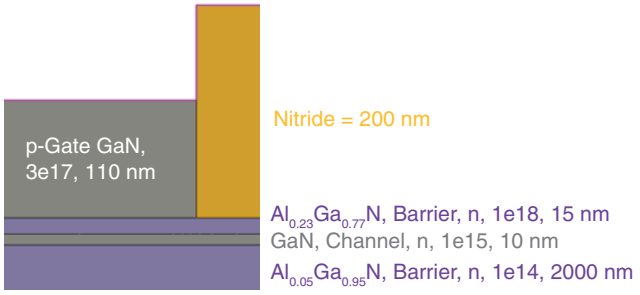


Figure 1 Cross section of simulated device structure (gate region)

The mesh for the GaN HFET device is generated by Sentaurus Mesh, which is called from within Sentaurus Structure Editor using:

```
(sde:build-mesh "" "n@node@_msh")
```

As for other heterostructure transistors, the mesh must be sufficiently tight around heterointerfaces, especially where large variations in carrier concentration are observed within short distances, for example, the 2D electron gas. This is accomplished using the `MaxLenInt` refinement function.

Device Simulation Using Sentaurus Device

Sentaurus Device simulates the DC characteristics of the p-gate GaN HFET. The curve information is stored in the `_des.plt` files of the corresponding Sentaurus Device instances (`BV`, `IdVg`, and `IdVd`).

AlGaN and GaN possess polarized wurtzite crystal structures, having dipoles across the crystal in the [0001] direction. In the absence of external fields, this macroscopic polarization includes spontaneous and strain-induced (piezoelectric) contributions. The primary effect of polarization is an interface charge (polarization charge) due to abrupt divergence in the polarization at the AlGaN–GaN heterointerface.

The spontaneous and piezoelectric polarization is simulated automatically by including the following in the `Physics` section:

```
Piezoelectric_Polarization(strain)
```

The $Al_{0.05}Ga_{0.95}N$ buffer is assumed to be completely relaxed, and the subsequent layers (channel and barrier) are strained to match the lattice constant of $Al_{0.05}Ga_{0.95}N$ but with 20% of relaxation. Therefore, `a0` and `relax` are set to the lattice constant of $Al_{0.05}Ga_{0.95}N$ and 0.2, respectively, in the `Piezoelectric_Polarization` section of the parameter files.

Moreover, the epitaxial layers are assumed to grow in the [0001] Ga-face direction by specifying the following settings in the parameter files:

```
LatticeParameters {
  X = (1, 0, 0)
  Y = (0, 0, -1)
}
```

The large polarization divergence at the AlGaN barrier surface (barrier–nitride interface) produces a huge sheet of negative polarization charge. This would accumulate holes at that interface and would completely deplete the channel of electrons. In reality, it is still unclear whether the polarization charge is compensated by fixed charges or interface trap states. In the project, they are completely compensated by deep, single-level trap states.

To match the subthreshold slope of the I_d–V_g curve in [1], traps are added to the AlGaN barrier–GaN channel interface.

It is also found that the passivation nitride thickness and the field-plate length impact the off-state leakage current and the breakdown voltage. Since they are not given in [1], guesses have been used.

Traps in the buffer are found to affect both the subthreshold slope and the off-state leakage current.

In large bandgap semiconductors, the intrinsic carrier density is extremely small. For this reason, it is recommended to adjust some default settings for numeric accuracy in the global `Math` section:

```
Math {
  ...
  ErrRef(electron)=1e5
  ErrRef(hole)=1e5
  RHSmin= 1e-15
  DirectCurrentComputation
  eDrForceRefDens= 1e8
  hDrForceRefDens= 1e8
  ...
}
```

The parameter `ErrRef(electron|hole)` defines to which carrier concentration level the accuracy of the solution variable effectively is controlled. The default value of 10^{10} cm^{-3} might be too high for nitrides. A stricter accuracy setting of 10^5 cm^{-3} or smaller sometimes helps convergence.

During Newton iterations, Sentaurus Device accepts a solution as having converged if either the weighted error estimate (L2-norm) becomes less than 1 or the value of the right-hand side (RHS) becomes smaller than a certain value, given by the parameter `RHSmin`. By default, the value of `RHSmin` is 10^{-5}. It is recommended to tighten this setting for nitrides, and the project uses 10^{-15} as an example.

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`DirectCurrentComputation` instructs Sentaurus Device to change the way it computes currents at electrodes. By default, Sentaurus Device uses 'doping wells' to compute currents. While the default method is the most accurate for devices where a clear doping well is associated with each electrode, for example, a MOS device, it might fail for an HFET device since the source, gate, and drain touch the same region without clear doping wells separating them. When `DirectCurrentComputation` is switched on, terminal currents are computed by integrating current densities for the nodes touching each electrode.

`eDrForceRefDens` and `hDrForceRefDens` are the numeric damping parameters for the driving field in high field saturation. By setting positive values, the convergence improves where strong generation–recombination occurs in regions with small carrier densities.

In the simulations, Schottky contacts are used in the source and drain for robustness. Only electron tunneling is switched on, and the electron tunneling mass is made to be arbitrarily small so the contact is essentially Ohmic. Hole tunneling is not switched on for simplicity. Such settings result in a better convergence rate than using Ohmic contacts, by avoiding abrupt band bending near the contacts. You also can tune the tunneling mass to model the contact resistance.

To limit the gate current to the experimental level [1] when the gate junction is forward biased, the p-gate GaN/contact is assumed to be Schottky, and the resistance is modeled using the Schottky resistance model to simplify the setup.

To improve the convergence of the breakdown simulation, transient simulation is used with minute amounts of background carrier generation. In addition, only electron-induced impact ionization is switched on (`eAvalanche`). This is because, in the breakdown regime, the electron impact ionization rate is much larger than the hole impact ionization rate.

Visualization With Sentaurus Visual

The last tool in the Sentaurus Workbench tool flow is the visualization tool Sentaurus Visual, which plots the I–V curves. The three Sentaurus Visual instances are:

- The first (`svisual_BV`) plots the off-state BV curve at Vg = 0 V.
- The second (`svisual_IdVg`) plots the I_d–V_g curves.
- The third (`svisual_IdVd`) plots the I_d–V_d curves.

Figure 2 to Figure 4 show the off-state BV curve of Structure 2 (`FP = 5` and `Lgd = 10`), and the I_d–V_g and I_d–V_d curves of Structure 1 (`FP = 1.8` and `Lgd = 6`).

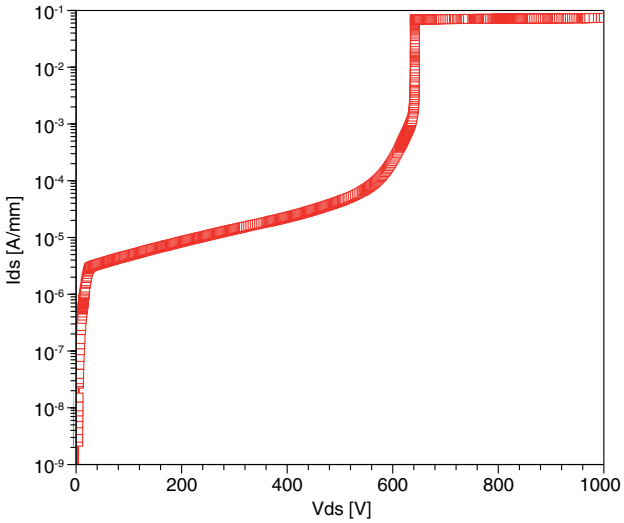


Figure 2 Off-state leakage curve of FP = 5 and Lgd = 10 structure

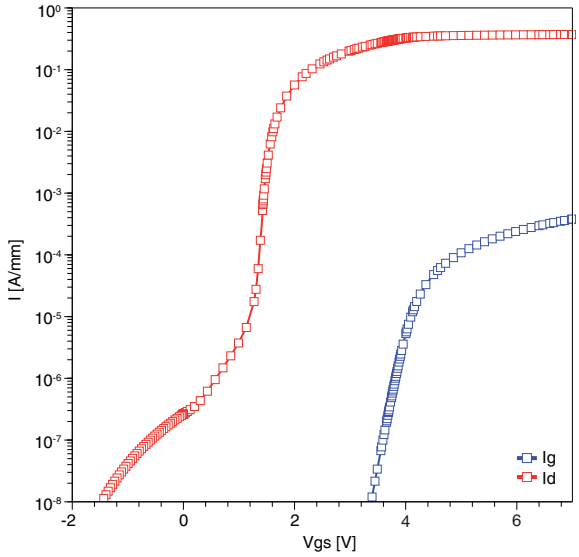


Figure 3 I_d–V_g and I_g–V_g curves of FP = 1.8 and Lgd = 6 structure with V_d = 15 V

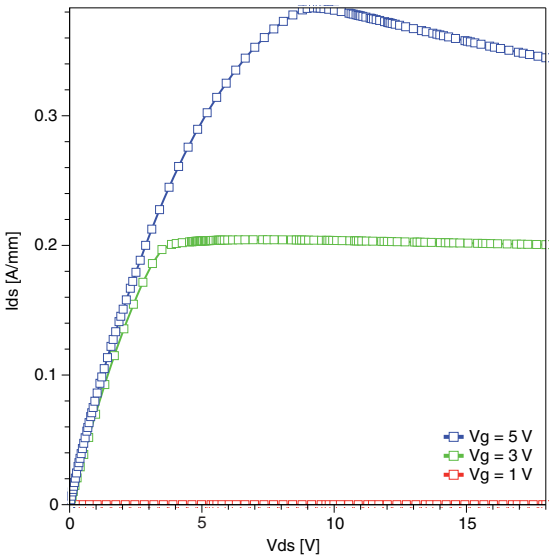


Figure 4 I_d–V_d curves of FP = 1.8 and Lgd = 6 structure

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References

[1] O. Hilt *et al*., "Normally-off AlGaN/GaN HFET with p-type GaN Gate and AlGaN Buffer," in *International Symposium on Power Semiconductor Devices and ICs (ISPSD)*, Hiroshima, Japan, pp. 347–350, June 2010.

[2] O. Ambacher *et al*., "Two-dimensional electron gases induced by spontaneous and piezoelectric polarization charges in N- and Ga-face AlGaN/GaN heterostructures," *Journal of Applied Physics*, vol. 85, no. 6, pp. 3222–3233, 1999.

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